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MD. HRIDOY HOSSEN

Date of birth: 30/06/2002 | **Place of birth:** PAKURIA, Bangladesh | **Nationality:** Bangladeshi | **Phone number:** (+880) 01775014937 (Mobile) | **Email address:** hridoy1922025.stat@std.iu.ac.bd | **Website:** https://github.com/hossenhridoy56 |

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ABOUT ME

Statistics graduate with a strong foundation in statistical analysis, statistical modeling, and machine learning. Experienced in applied research involving environmental and time-series data, with practical experience in clustering, data analysis, and statistical computing. Currently working as a Research Assistant on a UGC-funded project, co-authoring research submitted to Theoretical and Applied Climatology, and developing open-source statistical tools. Passionate about computational statistics, machine learning, and statistical methods for complex environmental and scientific data.

EDUCATION AND TRAINING

CURRENT Kushtia, Bangladesh
MASTER OF SCIENCE (M.SC.) (IN PROGRESS) Islamic University, Kushtia

Thesis-based programme | Expected completion: March 2027

Website https://www.iu.ac.bd/ | **Field of study** Statistics |

Thesis Model-Based Clustering of Climatological Functional Data with Temporal Dependence: An Autoregressive Mixture Model for Bangladesh Monsoon Rainfall

26/12/2019 – 26/11/2025 Kushtia, Bangladesh
BACHELOR OF SCIENCE (B.SC.) IN STATISTICS Islamic University, Bangladesh

Relevant areas of study: Statistical Inference, Regression Analysis, Time Series Analysis, Probability and Measure, Bayesian and Classical Inference, Matrix Algebra, Sampling Techniques, Multivariate Analysis, Stochastic Processes.

Website https://iu.ac.bd/ | **Field of study** Statistics | **Final grade** CGPA: 3.46 / 4.00 | **Level in EQF** EQF level 6 |

Type of credits Bangladeshi Credit System | **Number of credits** 160

2017 – 2019 Bangladesh
HIGHER SECONDARY CERTIFICATE (H.S.C.) Qadirabad Cantonment Sapper College, Natore

Field of study Science | **Final grade** GPA: 4.67 / 5.00 | **Level in EQF** EQF level 4

2017
SECONDARY SCHOOL CERTIFICATE (S.S.C.) Sara Marwari High School, Pabna

Field of study Science

RESEARCH EXPERIENCE

31/07/2025 – 30/06/2026
A Cluster Analysis of Bangladesh's Monthly Climate and Its Comparison with the Traditional Six-Season Calendar

UGC-Funded Research Project (Fiscal Year 2025-2026)
Manuscript: **A Cluster Analysis of Bangladesh's Monthly Climate and Its Comparison with the Traditional Six-Season Calendar**
Authors: Md. Matiur Rahman Molla, Md. Hridoy Hossen, Ayesha Siddika (*Submitted to Theoretical and Applied Climatology*, 2026; currently under peer review)

Supervisor: Md. Matiur Rahman Molla, Associate Professor, Department of Statistics & Data Science, Islamic University, Kushtia.

- Analyzed 30 years (1985–2014) of monthly climate observations comprising 10,865 station-months across 33 BMD weather stations in a 6-variable climate space (Tmax, Tmin, rainfall, humidity, wind speed, cloud cover).
- Applied K-means, DBSCAN, and hierarchical agglomerative clustering along with PCA and Dynamic Time Warping (DTW) to discover robust seasonal boundaries.
- Found empirical climatological evidence supporting a four-season structure (Winter, Pre-Monsoon, Monsoon, Post-Monsoon) with October as an independent transition month, challenging the traditional six-season paradigm.
- Published complete open-access analysis pipeline, cleaned datasets, and reproducible scripts on GitHub and Zenodo.

Links https://github.com/hossenhridoy56/Bangladesh-Seasonal-Reclassification | https://doi.org/10.5281/zenodo.23324622

Hydro-Climatic Research Project: Resolving Monsoon Timing Shifts in Bangladesh (1981–2025): Observational Uncertainty, Autoregressive Bayesian Model Selection, and Finite-Sample Detection Limits

Research Project & Manuscript Prepared for Journal Submission
Supervisor: Md. Matiur Rahman Molla, Associate Professor, Department of Statistics & Data Science, Islamic University, Kushtia.

- Analyzed 45 years of precipitation data from ERA5-Land and CHIRPS datasets across Bangladesh.
- Developed rainfall-based indices to assess monsoon onset, peak, withdrawal, and seasonal variability.
- Applied trend detection techniques, including Mann-Kendall tests and False Discovery Rate (FDR) correction.
- Implemented Bayesian time-series models incorporating ENSO variability to evaluate monsoon timing shifts.
- Conducted Monte Carlo simulations to assess trend detectability and statistical uncertainty.

CURRENT

MVT Project: A Novel Massive Variance Stability Test: Theory, Distributional Derivations, and Superior Performance Under Structural Breaks

Research Manuscript & Open-Source Software Package (Manuscript in Preparation)

- Formulated the Massive Variance Stability Test (MVT)—a global test statistic designed to evaluate variance stationarity across time series by partitioning observations into non-overlapping sub-segments.
- Derived exact finite-sample distributional representations and asymptotic behavior under the Gaussian homoskedastic null hypothesis.
- Executed comprehensive Monte Carlo simulations evaluating empirical size distortion, detection power, and robust performance under structural breaks, heteroscedasticity, and autoregressive persistence.
- Developed and maintained open-source simulation engines and figure-generation packages in Python and R.

Link https://github.com/hossenhridoy56/massive-variance-stability-test

Climate-Health Vulnerability Assessment in Bangladesh: A Data-Driven Statistical Analysis

- Developed a district-level climate-health vulnerability framework for 64 districts of Bangladesh using ERA5-Land reanalysis data (1995–2024) and MICS 2025 health, socioeconomic, and disaster indicators.
- Applied Principal Component Analysis (PCA) to seven climate indicators, extracting three components explaining 89.73% of total variance.
- Constructed standardized composite indices for Climate Exposure, Health Sensitivity, Adaptive Capacity, and Disaster Exposure.
- Identified substantial spatial heterogeneity, with coastal and hill-tract districts (Bhola, Patuakhali, Bandarban, Rangamati, and Cox's Bazar) exhibiting acute composite vulnerability.

05/2025 – CURRENT

B.Sc. (Hons.) Project – Intelligent Departmental Gateway with Generative AI

- Developed a web-based departmental management system with Generative AI–powered document automation.
- Automated academic document generation (transcripts, certificates, attendance reports).
- Built and deployed a full-stack application using Python (Flask), SQLAlchemy, PostgreSQL, and Bootstrap.
- Implemented role-based dashboards and conducted performance testing (Locust: 24.4 RPS, 0% failure rate).

Links https://stratosphere-gateway.onrender.com/ | https://github.com/hossenhridoy56/stratosphere-gateway

SKILLS

Statistical & Data Analysis

R | Python | SAS language | SPSS | STATA | SQL | Clustering (K-Means, DBScan, GMM, Hierarchical) | Machine Learning | Bayesian Inference & Time Series Modeling | Exploratory Data Analysis & Monte Carlo Simulation

Data Handling, Reporting & Automation

LaTeX | Microsoft Office (Word, PowerPoint, Excel) | Git & GitHub

Programming & Data Tools

PostgreSQL | SQLAlchemy | Flask & Web Scraping | Python Scientific Stack (NumPy, Pandas, SciPy, Scikit-learn, Xarray)

WORK EXPERIENCE

ASSISTANT TEACHER (SCIENCE & MATHEMATICS) – EDUCARE IDEAL SCHOOL & COLLEGE – 26/07/2024 – Current – BANGLADESH

Business or Sector: Education

- Taught Mathematics, Physics, and Chemistry at SSC level.
- Developed lesson plans, assessments, and examination materials.
- Evaluated student performance and provided academic support.
- Conducted tutorial sessions and assisted with laboratory activities.
- Maintained academic records and monitored student progress.

EXTRACURRICULAR ACTIVITIES & MEMBERSHIPS

Member, Islamic University Model United Nations Association (IUMUNA)

- Active member participating in diplomatic simulations, debate on global socio-economic and climate policy issues, public speaking, negotiation, and collaborative resolution drafting.

Chess Champion, Annual Sports Week

Department of Statistics & Data Science, Islamic University, Kushtia

- Champion in the competitive departmental chess tournament during annual sports week, demonstrating advanced strategic calculation, foresight, and tactical decision-making under time pressure.

TRAINING & CERTIFICATIONS

11/2024 – 01/2025

Basic Programming with Python (60 hours)

Department of Information and Communication Technology (ICT),
Islamic University, Bangladesh
(EDGE Project – Bangladesh Computer Council, ICT Division)

- Python fundamentals, data structures, basic data handling, introductory use of standard libraries (NumPy, Pandas), and basic exposure to Flask for web applications.

Basics of Artificial Intelligence

Robi Axiata Limited, powered by 10 Minute School
Issued: March 2022.

- Foundational understanding of artificial intelligence, machine learning concepts, and data-driven decision-making, with introductory exposure to AI applications and automation.

23/05/2023 – 25/05/2023

C Programming Language Workshop

Islamic University IT Society (IUITS)
Islamic University, Kushtia, Bangladesh

Workshop Coverage:

- Fundamentals of C programming, control structures, and basic problem-solving using structured programming concepts

LANGUAGE SKILLS

Mother tongue(s): **BANGLA**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH — IELTS ACADEMIC: 7.0	C1	C1	B2	B2	B2